

Contribution report

FairTox — Parity, Utility and Safety Are Not Independent

DLE-AI-202, Cohort I 2026 · Track 2 · <https://github.com/nigarrustamova/fairtox> (tag v1.0-final)

Member	Code	Report sections	Experiments	Slides	Other
Aliyev, Orkhan <code>@mrorxan</code>	Stage registry and scaffolding; loss weighting and Group DRO (<code>training/losses.py</code>); comparability and reuse fingerprints (<code>provenance.py</code>); benchmark bias metrics (<code>fairness/jigsaw_metrics.py</code>); controlled comparison (<code>stages/compare.py</code>); ablation (<code>stages/ablate.py</code>); mechanism analysis (<code>stages/mechanism.py</code>); cross-family statistics (<code>evaluation/aggregate.py</code> , <code>scripts/summarise_study.py</code>); run orchestration (<code>pipeline.py</code>)	Method; Results A–D, F–H (headline pair, all eight configurations, mechanism, threshold matching, Group DRO, held-out axes, ablation); Discussion; Conclusion	Headline pair over eight seeds; fixed-epoch control; Group DRO; held-out-axis arm; α sweep and the <code>class</code> control; mechanism and matched-threshold analyses	15–19 (mechanism, the loss, headline, all families, the three controls)	Experiment plan and pre-registered outcomes; scheduling of the booked GPU window
Mirzayeva, Laman <code>@lmmnrzyv</code>	Global classification metrics (<code>evaluation/metrics.py</code>); stratified split and fingerprint (<code>data/split.py</code>); subgroup auditor with Wilson intervals and support floors (<code>fairness/auditor.py</code>); all plotting (<code>utils/plotting.py</code>); audit stage; error analysis and the coded sample; per-subgroup thresholds (<code>stages/postproc.py</code>); duplicate check (<code>stages/leakage.py</code>); optional analyses; CI workflow	Problem Formulation (two errors, three axes, the three guards, pre-registration); Results I–M (threshold-free metrics, post-processing, the manual coding, who is affected, integrity and leakage); Limitations	Subgroup audits and threshold sensitivity; the training-free post-processing arm; the leakage measurement; manual coding of 60 identity-bearing false positives	7–10 (three axes, five guards, what the errors are, two coded examples)	Figure review before submission; test coverage for two other members' modules
Rustamova, Nigar <code>@nigarrustamova</code>	Repository, branch protection and release tag; logging (<code>utils/logging_utils.py</code>); configuration system (<code>config.py</code> , <code>configs/default.yaml</code> , <code>smoke.yaml</code>); artefact I/O and prediction format (<code>utils/io.py</code> , <code>evaluation/predictions.py</code>); corpus loader and synthetic corpus (<code>data/load.py</code>); EDA and TF-IDF baseline stages; command-line entry point (<code>scripts/run_all.py</code> , <code>run_all.sh</code>); README.md	Introduction; Related Work (measuring unintended bias, and the gap this paper fills); Experimental Setup (corpus and splits, implementation and reproducibility); Tools and Acknowledgements	Corpus EDA and subgroup support counts; the TF-IDF + logistic-regression baseline; the synthetic smoke pipeline used to validate the code path	3–6 (the problem, the three intervention points, the corpus, the classical baseline)	Repository owner and submitter; one-command reproduction path; documentation review
Samadov, Nijat <code>@NijatSamadov</code>	Device and seed utilities; tokenisation and dataset (<code>data/dataset.py</code>); classifier head (<code>models/transformer.py</code>); fine-tuning loop (<code>training/trainer.py</code>); training stage (<code>stages/train.py</code>); counterfactual augmentation (<code>data/augment.py</code>); token attribution (<code>explain/attributions.py</code>); the eight booked-window configuration files; TRAINING_DAY.md and the Kaggle runner notebook	Related Work (pre-processing); Experimental Setup (model and training, compute); Results E (counterfactual augmentation, dose response)	All 115 trainings inside the booked window; the DistilRoBERTa backbone arm; counterfactual swapping at both doses; the five-run noise-floor measurement	11–14 (the model, 115 models and the noise floor, the swap, the null result)	Server environment (MIG targeting, CUDA build); recovery of the interrupted window

Short description per member

Aliyev, Orkhan	Owned the objective and the comparison machinery: the per-sample weighting, the Group DRO control, the fingerprints that make a controlled pair legitimate, and the stages that turn two arms into a verdict. Wrote the mechanism analysis — the regression of subgroup FPR on subgroup toxic rate — and the cross-family aggregation that produces every table in the paper. Ran the eight-seed headline replication and the three controls behind the paper's central claim.
Mirzayeva, Laman	Owned measurement. Wrote the metrics, the stratified split, and the subgroup auditor, including the two decisions that keep the fairness numbers honest: separate support floors for FPR and FNR, and Wilson intervals rather than a bootstrap. Wrote every figure routine, the error-analysis stage, and the training-free post-processing arm, and read and coded all 60 sampled false positives by hand.
Rustamova, Nigar	Set up the repository and the infrastructure the rest of the work sits on: configuration, logging, artefact I/O, the prediction format, the corpus loader and the driver that makes the study one command from a clean checkout. Wrote the EDA and the classical TF-IDF baseline the deep model is measured against, and the README. Submits the deliverables on behalf of the team.
Samadov, Nijat	Owned the model and the training path: tokenisation, the classifier head, the fine-tuning loop and the training stage, plus the configuration files for the booked window. Ran the window itself — 38 runs, 115 models, 34.4 GPU-hours — and diagnosed the CUDA and MIG problems that blocked it. Wrote the counterfactual-augmentation arm and the token-attribution analysis.

Balance and cross-review. No directory is owned by one person: `stages/` and `tests/` are shared by all four, and `utils/`, `data/`, `evaluation/`, `fairness/` and `training/` each have two or three owners. Tests are deliberately written by someone other than the module's author in eight pull requests, so every member has read and tested at least two other members' code. Every member reviewed pull requests from every other member.

Agreement. All four members have read this report and agree that it describes the division of work accurately. Prepared for the final submission of DLE-AI-202, September 2026.